Worksheet: Material balances in liquid-liquid extraction

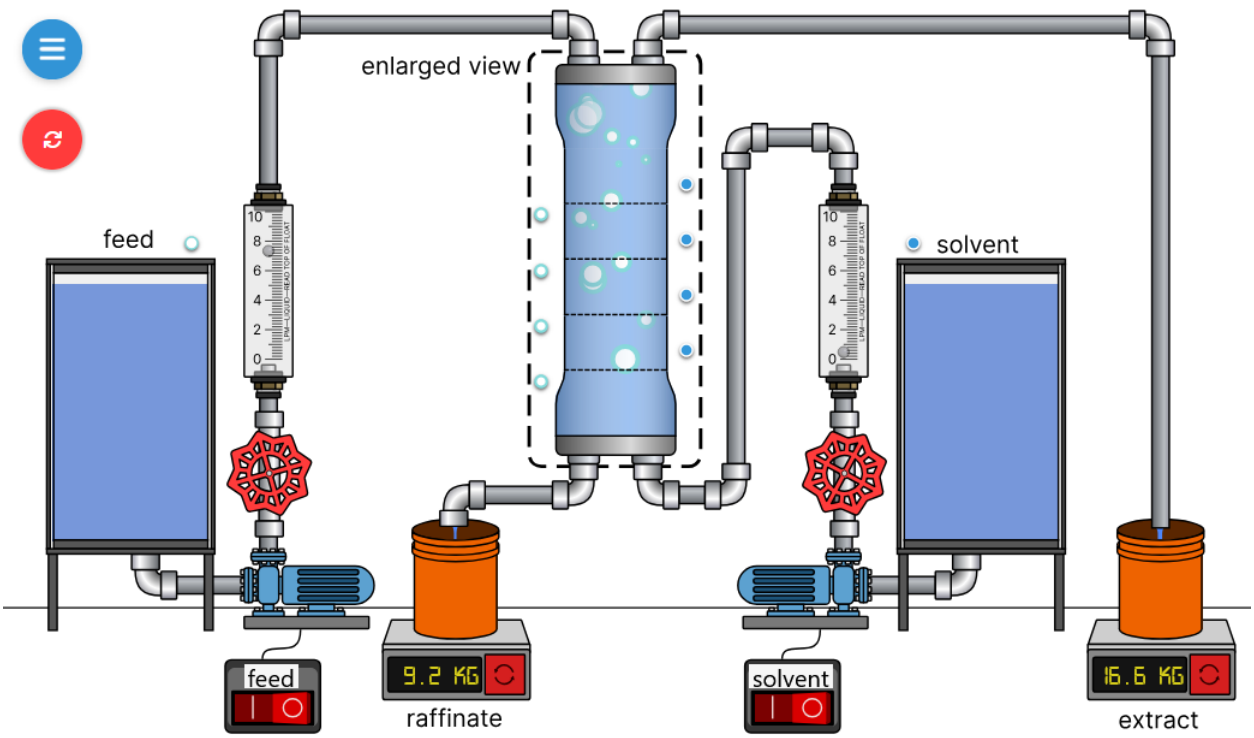
Name(s) _____

This experiment applies material balances to a liquid-liquid extraction process in which water as a solvent contacts an acetic acid/chloroform mixture in a steady-state counter flow experiment. Because acetic acid is much more soluble in water, it is extracted from the chloroform organic phase into the aqueous phase.

Student learning objectives

1. Be able to apply mass balances on a three-component system.
2. Be able to use phase equilibrium data to partially generate the phase envelope on a ternary phase diagram.
3. Be able to explain what happens during liquid-liquid extraction.

Equipment



The number of stages can be varied between three and five. In each stage, the aqueous solvent phase contacts the organic phase (acetic acid, chloroform, and a small concentration of water) in counter-current flow. The organic phase flows down and the solvent phase flows up. The phases leaving each stage are in phase equilibrium. The water extracts acetic acid solute from the organic feed. Thus, the extract contains the water solvent with acetic acid and some chloroform. Chloroform is enriched in the raffinate because acetic acid and water have been removed from it.

S = solvent (blue) F= feed (white) E = Extract R = Raffinate

Note that the feed and solvent tanks are large relative to the size of the column (thus the column is enlarged in the simulation view). Thus, the tanks volumes do not change significantly during an experiment.

Questions to answer before starting the experiment

1. How many independent mass balances can you write for each stage of this liquid-liquid extraction? Explain.

2. The two phases contact each other in a counter-current flow. What might be the advantage of counter-current flow over co-current flow?

3. Would you expect the feed stream (chloroform, acetic acid, water) to increase or decrease its flow rate as it flows down the extraction system? Why?

4. Would you expect the water solvent stream to increase or decrease its flow rate as it flows up through the extraction system? Why?

Before starting

Select the edit internals button just above the feed tank to choose the number of stages and record below.

Number of stages _____

Procedure:

1. Turn on the solvent pump by clicking the 1/0 switch and turn the solvent valve counterclockwise to start solvent flow into the extraction system. The column will fill with solvent. Once the column fills with solvent, turn on the feed pump (only possible after the column fills with solvent), and turn the feed valve counterclockwise to increase the feed flow rate. Record the two volumetric flow rates in Table 1.
2. The mass percentages of the three components in the feed and their densities are:

Component	Mass percent	Density (kg/L)
Acetic acid	52.0	1.49
Chloroform	43.0	1.04
Water	5.0	1.00

The solvent only contains water.

3. The density of a mixture is $\rho_{mix} = 1/\sum(x_i/\rho_i)$ where x_i is the mass fraction and ρ_i is the density of component *i*. Calculate fluid density in the feed tank and record it in Table 1.

4. Use the volumetric flow rates and densities to calculate the mass flow rates of the feed and the solvent and record the values in Table 1.
5. Allow time for the system to reach steady state and liquid to accumulate in the extract and raffinate tanks. Then determine the mass flow rates of the two exit streams by recording the mass change in the buckets while recording the time elapsed with your phone. Record values in Table 1.

Table 1

Tank	Volumetric flow rate (L/min)	Density (kg/L)	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Feed					
Solvent		1.00			
Extract					
Raffinate					

5. Repeat measurements of the extract and raffinate mass flow rates and record in Table 2.

Table 2

Tank	Elapsed time (s)	Mass change (kg)	Mass flow rate (kg/min)
Extract			
Raffinate			

6. Calculate average extract and raffinate mass flow rates and record in Table 2. Also copy the feed and solvent mass flow rates from Table 1 into Table 3.

Table 3

Tank	Mass flow rate (g/s)
Feed	
Solvent	
Extract	
Raffinate	

7. Measure the compositions of each of the streams by moving your mouse over the dots on the slides of the column. The top stage in the column is stage 1. The aqueous phase leaving stage 1 is stream E_1 and the organic phase leaving stage 1 is R_1. Similarly, the other E_i and R_i streams refer to the streams leaving stage i.

Table 4

	Mass fractions		
Stream	Acetic acid	Chloroform	Water
E_1			
E_2			
E_3			
E_4			
E_5			
S			
F			
R_1			
R_2			
R_3			
R_4			
R_5			

Data Analysis:

Do an overall mass balance on the system. That is, add the mass flow rates of feed and solvent into the extraction column and enter the value in the Total mass balance row in Table 5. Add the mass flow rates of raffinate and extract streams out of the column and enter that value in the Total mass balance row in Table 5. Calculate the differences and the percentage differences and enter those values in Table 5. Likewise, calculate the total flow rates in and out of each of the species and enter those values in the corresponding rows in Table 5. Calculate the differences and the percent difference and enter those values into Table 5.

Table 5

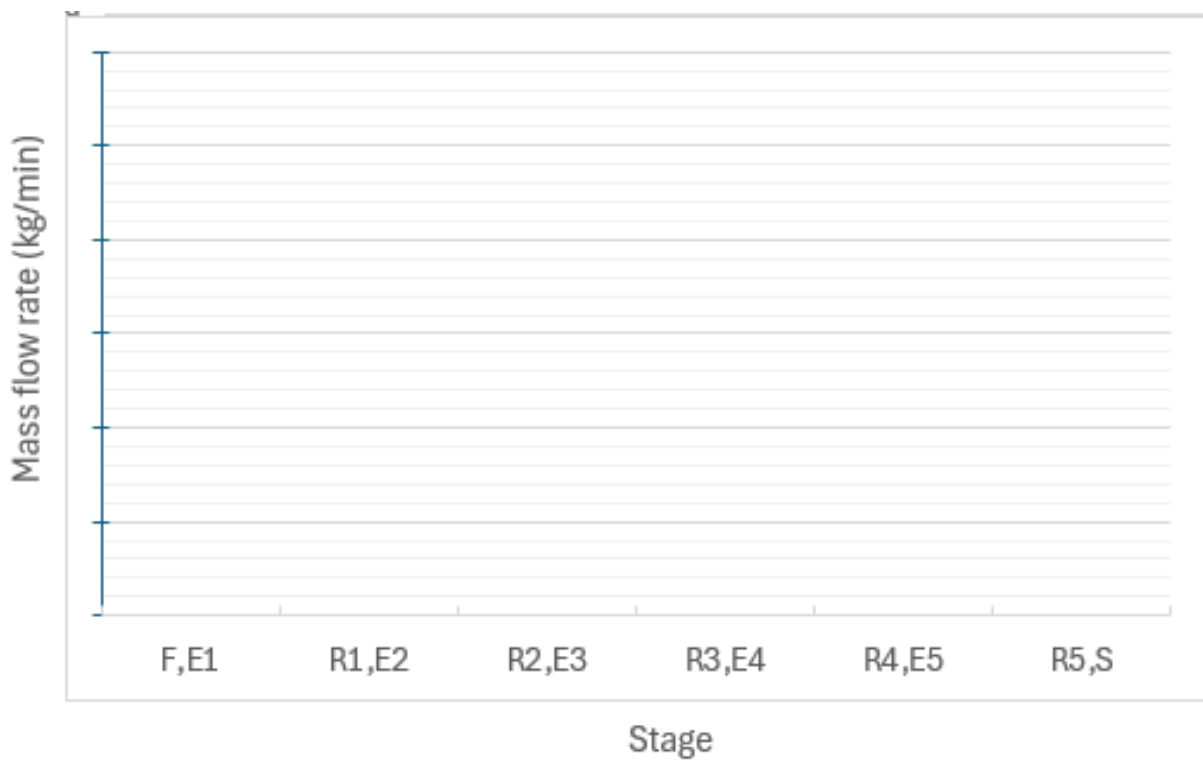
	Flow in (kg/min)	Flow out (kg/min)	Difference: in-out (kg/min)	Percent difference
Total mass balance				
Acetic acid				
Chloroform				
Water				

Carry out component mass balance for each stage to determine the total flow rates into and out of each stage and enter those values in Table 6.

Table 6

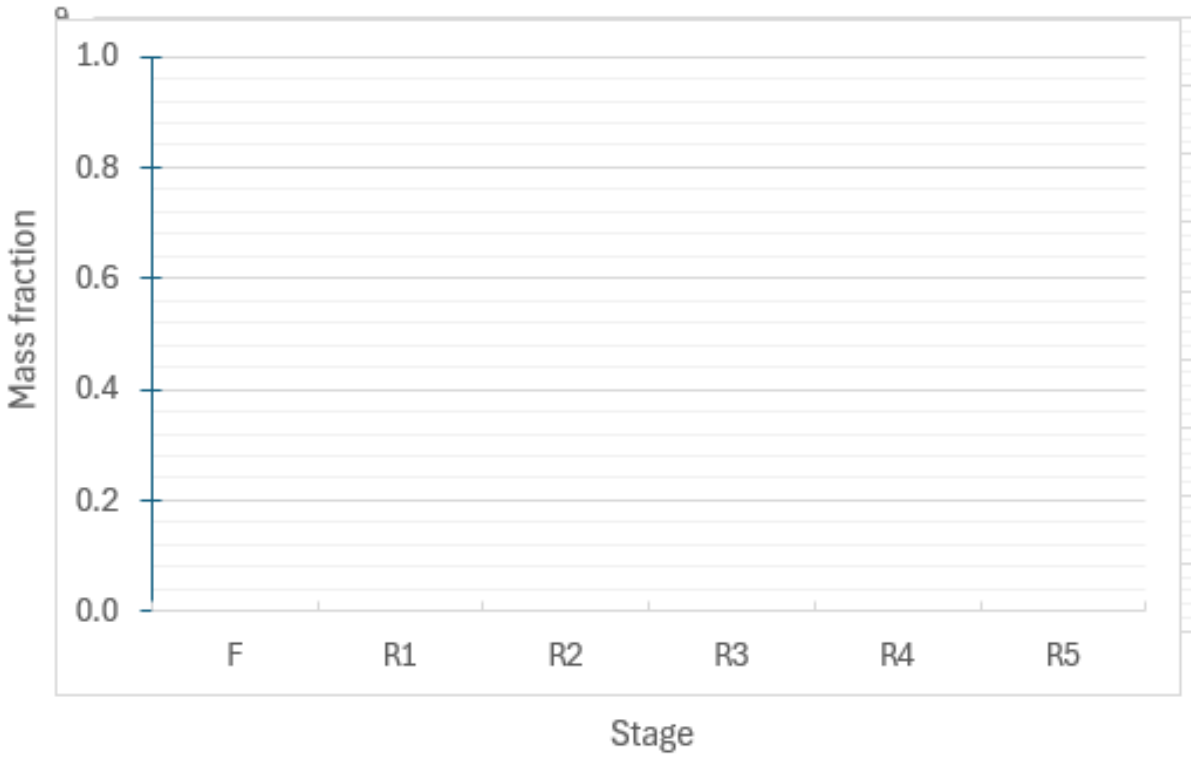
Stream	Mass flow rate (g/s)
R_1	
R_2	
R_3	
R_4	
R_5=raffinate	
E_1=extract	
E_2	
E_3	
E_4	
E_5	

Plot mass flow rates of the streams on the y-axis versus the stage as indicated in the figure below.



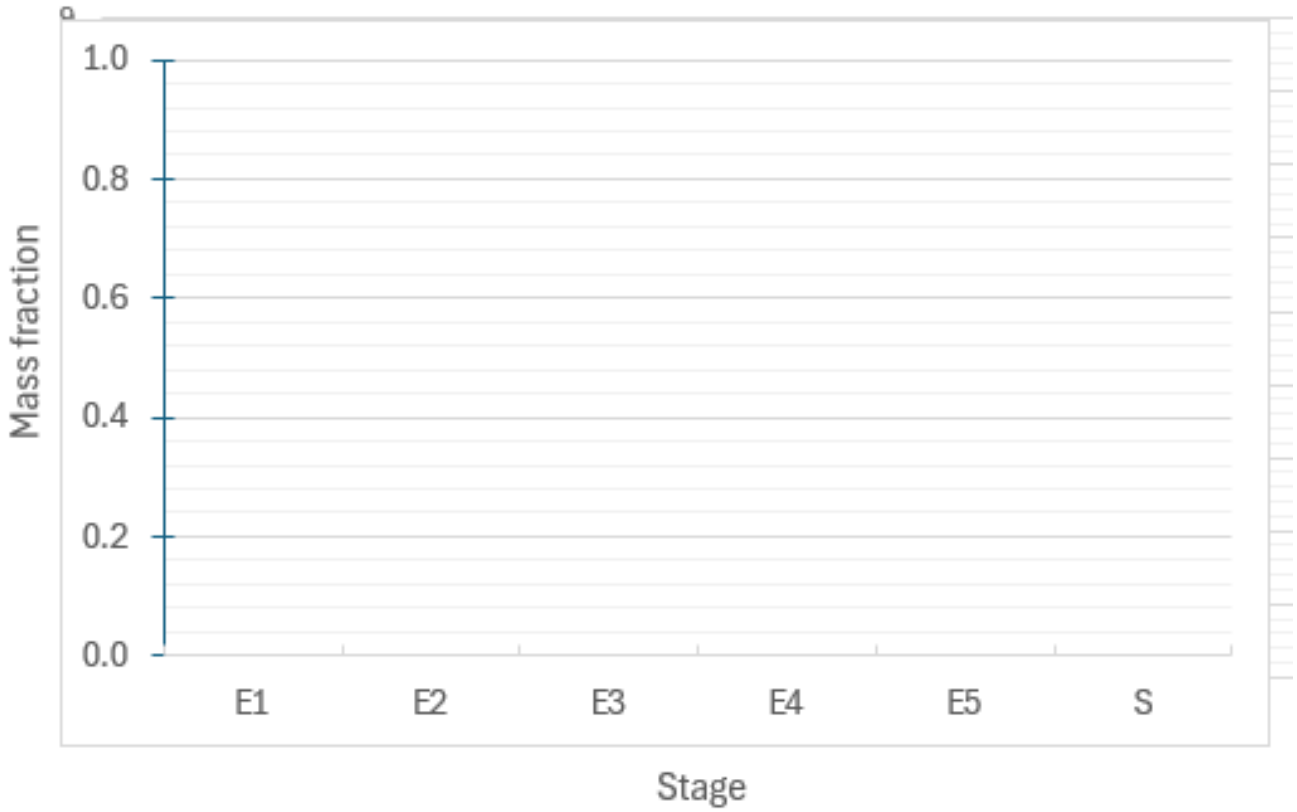
Explain the behavior in the graph.

Plot the mass fractions of <u>each</u> component for the feed flow stream.



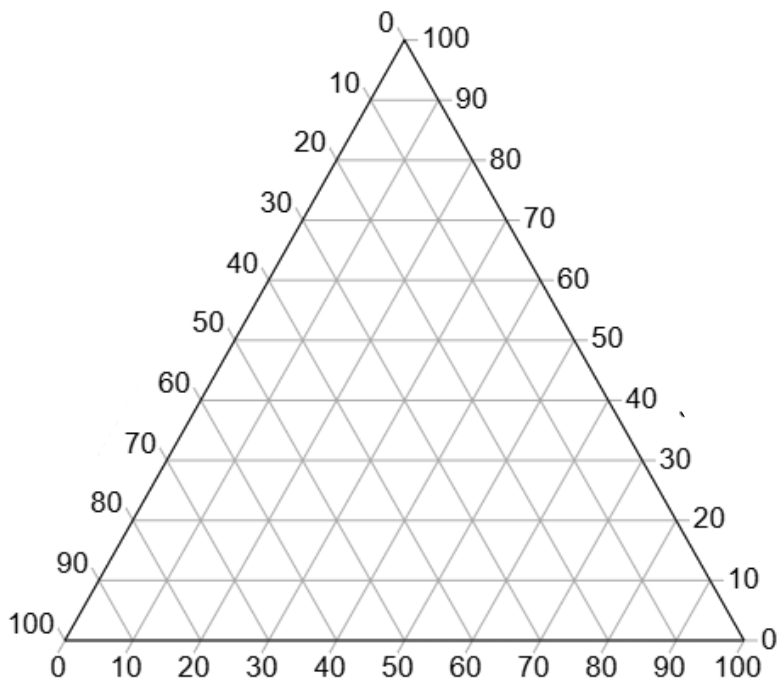
Explain the behavior in the graph.

Plot the mass fraction of each component for the solvent flow stream.



Explain the behavior in the graph.

The streams leaving each stage (e.g., E_2 and R_2) are in equilibrium. Tie lines connect two composition points that are in equilibrium on a ternary phase diagram. Points connected by tie lines are on the phase envelope of a ternary phase diagram. Locate the points (exit stream concentrations) on the ternary phase diagram below and draw as much of the phase envelope as you can. Also draw the tie lines.



Questions to answer

1. Where might these measurements have errors?

2. What safety precautions would you take to conduct this experiment in the laboratory?